



Citrus yield estimation for individual trees integrating pruning intensity and image views

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ARTICLE INFO

Keywords:

Citrus yield
Image detection
Yield estimation
Individual trees
Pruning

ABSTRACT

Accurately estimating the yield of citrus fruit on individual trees is essential for precise orchard management and the income of producers. However, estimating the yield of citrus fruit from images of trees remains challenging among different processes of tree pruning and image acquisition. This study adopted a deep learning based detection model to count fruit in tree images and machine learning models to estimate the yield of individual trees from the fruit count. Trees under four levels of pruning intensity (no pruning, 0–5 %, 5–10 %, and 10–15 % of new sprouts pruned) and imaged from three different views (two, four, and six images per tree) to determine the optimal conditions for yield estimation. The variables considered for yield estimation included fruit count, pruning intensity and image views. Dataset containing 1200 tree images were used to train and test four machine learning models: random forest, support vector machine, extreme gradient boosting (XGBoost), and generalized linear model. The XGBoost model achieved the lowest errors in both training and testing. The optimal yield estimation occurs when there are two, four, and six image views and trees that have been pruned >10 %, 5–10 %, and ≤5 %, respectively. The findings can enhance the accuracy of image based citrus fruit yield estimation for individual trees and reveal the influences of pruning and image views.

1. Introduction

Citrus is the world's most commonly eaten fruit and one of the most important economic fruit crops, with yield and cultivation area increasing every year. Increasing fruit yield and having an accurate yield estimation is essential to generating income for the producers (Costa et al., 2022). A detailed estimation of the number of fruit on individual trees would also promote precision management in citrus orchards (Xin et al., 2022). For this specific task, deep learning-based image target detection models offer more efficient and accurate assistance in the field of harvest of citrus fruit and the estimation of orchard yields (Ojo et al., 2021). These models have considerably helped the management of citrus orchards, but the accuracy of estimating fruit yield still varies widely (Chen et al., 2024). Thus, the challenge with image-based fruit yield estimating is to comprehend the relationship between the fruit

recognized from the tree images and the actual fruit on the trees (Apolo-Apolo et al., 2020). In this regard, research has suggested that only 70–85 % of the fruit is present in the tree canopy images and this proportion could be influenced by canopy architecture, pruning, and camera views (Gongal et al., 2015). Therefore, it is important to study the influence on the recognition of image-based citrus fruit, so the fruit presented in images can be accurately detected.

Pruning is a significant factor that influences the estimation of citrus fruit yield based on images. Pruning usually take place in the winter and spring to promote the new growth of leaves and twigs, ensuring optimal nutrient distribution to newly emerging flowers and fruits (Liu et al., 2020). Pruning with appropriate intensity also removes diseased/weak shoots and increases canopy light use efficiency (Budiarto et al., 2022). However, different pruning intensities alter the proportion of occluded fruits in images, posing a significant challenge for fruit yield estimation

Abbreviations: ANOVA, analysis of variance; CSP, Center and Scale Prediction; EasyDAM, Easy domain adaptation method; FOV, field of view; FPN, feature pyramid network; GLM, generalized linear model; MAPE, mean absolute percentage error; PAN, Path Aggregation Network; PE, percentage error; POLY, polynomial function; RBF, radial based function; R-CNN, Regions-convolutional neural network; RF, random forest; RoI, Region of Interest; RPN, region proposal network; SMAPE, symmetric mean absolute percentage error; SVM, support vector machine; UAV, unmanned aerial vehicles; XGBoost, extreme gradient boosting; YOLO, you only look once.

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<https://doi.org/10.1016/j.eja.2024.127349>

Received 17 May 2024; Received in revised form 5 September 2024; Accepted 5 September 2024

Available online 16 September 2024

1161-0301/© 2024 Published by Elsevier B.V.

based on images (Wang and He, 2021). Therefore, fruit yield estimation for individual trees based on images should address the influence of different pruning intensities.

The image acquisition process also affects the accuracy of citrus fruit yield estimation, particularly when detailed into individual trees. There are mainly two types of image acquisition methods: aerial imaging (Huang et al., 2022) and ground imaging (Zhang, Wang et al., 2022). Aerial imaging, usually based on unmanned aerial vehicles (UAV), is faster than ground imaging and applicable to both flat ground and mountain orchards (Moriya et al., 2021). Ground imaging, on the other hand, acquire images from more perspectives and locate the view of individual trees (Radcliffe et al., 2018). It is also applicable to orchard blocks with greenhouses, where aerial imaging may face occlusion issues. The estimation of citrus fruit yield based on both image acquisition methods relies on deep learning target detection models, such as Faster R-CNN (region-convolutional neural network) (Gao et al., 2020) and YOLO (you only look once) (Zheng et al., 2021). In terms of fruit counting and yield estimation, Faster R-CNN tends to be more precise and suitable for detection with complex background (Mai et al., 2020), while YOLO is lightweight and responsive with acceptable accuracy (Xu et al., 2023).

The processes of pruning and acquiring images to estimate the yield of citrus fruit of individual trees by counting fruit remains a challenging task. Deep learning-based methods, particularly convolutional neural networks (CNNs), have been widely adopted for various agricultural applications because of its intuition and accuracy (Khattak et al., 2021). For instance, Mazzia et al. (2020) employed a CNN for estimating apple yield, achieving significant accuracy improvements over traditional methods. Similarly, Karydas et al. (2020) applied deep learning techniques to estimate cherry yield, highlighting the model's ability to handle varying lighting conditions and occlusions in images. Zeeshan et al. (2023) demonstrated the effectiveness of deep learning models in detecting citrus fruits, emphasizing the importance of model robustness across different growth stages and environments. de Luna et al. (2020) focused on tomato yield estimation using a combination of image processing and deep learning, underscoring the challenges posed by dense foliage and fruit clustering.

Despite the unique non-linear feature extraction approach makes deep learning models promising in fruit detection (Chen et al., 2021), several challenges remain in the domain of deep learning-based fruit detection and yield estimation. These challenges include handling occlusion and overlapping fruits in the images (Lee et al., 2020), as well as addressing the impact of various orchard management practices, such as pruning. These impact on the accuracy of deep learning models has not been thoroughly investigated (Matias et al., 2023).

Therefore, in this study, a YOLOv6 based model to detect citrus fruit was trained and tested on a self-collected citrus trees image dataset with four different levels of pruning intensity and three different imaging views. Machine learning models were constructed to correlate image-based detections with actual fruit presence on trees by validating the output against the manually counted fruits (ground truth). Finally, we proposed the optimal machine learning model to estimate the yield of citrus fruit for individual trees. The innovation of our proposed method lies in the comprehensive approach to improve robustness against occlusion and overlapping fruits by integrating YOLOv6 with machine learning models and optimizing image acquisition strategies. Additionally, we systematically evaluate the impact of pruning intensity on model performance, providing actionable insights for orchard management practices.

2. Materials and methods

2.1. Citrus trees and study location

Citrus cultivar Miyagawa (*Citrus unshiu* Marc. 'Miyagawa wase') was selected as the dataset. This cultivar is widely cultivated in southern

China. All the citrus trees were trimmed into similar branch architecture and cultivated in the orchard of the Institute of Citrus Fruit, Zhejiang Academy of Agricultural Sciences (Taizhou, China) (28.640 N, 121.165 E). The citrus trees for different pruning treatments were planted in separate orchard blocks more than five years ago. These orchard blocks were more than 50 m apart. The tree height and width were measured and recorded for each tree.

2.2. Pruning treatment and fruit counting

Pruning was conducted in the spring after the first flush of growth. The intensity of pruning treatments was designed as follows: 0 (no pruning), 0–5 %, 5–10 % and 10–15 % of the new sprouts of each tree were pruned. Each level of pruning intensity was performed on 25 citrus trees in the same orchard block as replicates. Thus, 100 trees were involved in the whole experiment.

Since citrus trees are planted in rows as shown in Fig. 1 A, a ground imaging system is usually used to acquire images of each tree. Thus, three types of image views (Fig. 1 A) were designed to test the optimal image acquisition strategy, which aims to maximize the visibility of fruits and minimizes occlusion.

- (1) Two views: This is the most common approach utilized by aerial and ground imaging. Two images were taken from the opposite side (180° rotation) of each citrus tree, perpendicular to the row direction;
- (2) Four views: An image was taken at every 90° rotation for four images from each citrus tree, two of which had angle views perpendicular to the row direction;
- (3) Six views: An image was taken at each 60° rotation for six images from each citrus tree, two of which had angle views perpendicular to the row direction.

The ground sampling distance (distance between the camera and the tree canopy) was kept the same (approximately 0.75 m) in all experiments, so that the field of view (FOV) and image size were also kept the same (approximately 77.38° and 1216 × 2160 pixels, respectively). The image direction was kept portrait to avoid covering fruits of other trees in the image. The percentages of tree width in the images under different pruning treatments were calculated and summarized in Table 1. Percentages greater than 100 % indicated the tree width covered the whole view.

A total of 1200 citrus tree images (taken from trees among 4 pruning intensity × 25 trees × (2+4+6) images per tree) were selected to form a dataset of fruit images. In most cases the fruits in the background were quite smaller than the fruits on the foreground tree. Thus, only the fruits on the foreground tree were labeled, which would prompt the fruit detection model less possible to detect fruits in the background. Easy-DAM_V2 (Zhang, Chen et al., 2022) was used to label the images of citrus fruit and conduct data augmentation when building the dataset of fruit images. The dataset was then randomly divided into a training set (70 %), and testing set (30 %). Data augmentation was conducted on the training set, including rotation of images by 90°, 180°, and 270°; horizontal and vertical mirroring; resizing 0.25, 0.5, 0.75, 2 and 4 times with bilinear interpolation; and Gaussian blurry with 5 × 5 and 10 × 10 kernel. After augmentation, the training set contained 100,800 images with labels.

The Faster R-CNN based fruit detection model was trained and tested on our dataset mentioned above. An FPN (feature pyramid network) was implemented to encode color and texture features (Apolo-Apolo et al., 2020). An improved RPN (region proposal network) was utilized to generate regional proposals based on the features and align the RoIs (Region of Interest). The model was then constructed by feeding the RoIs to the convolution layers for detection (Zhu, Gu et al., 2022). A total of 88 epochs were trained on the dataset of fruit images with batch sizes of two and the sum of bounding box regression loss as the loss function.

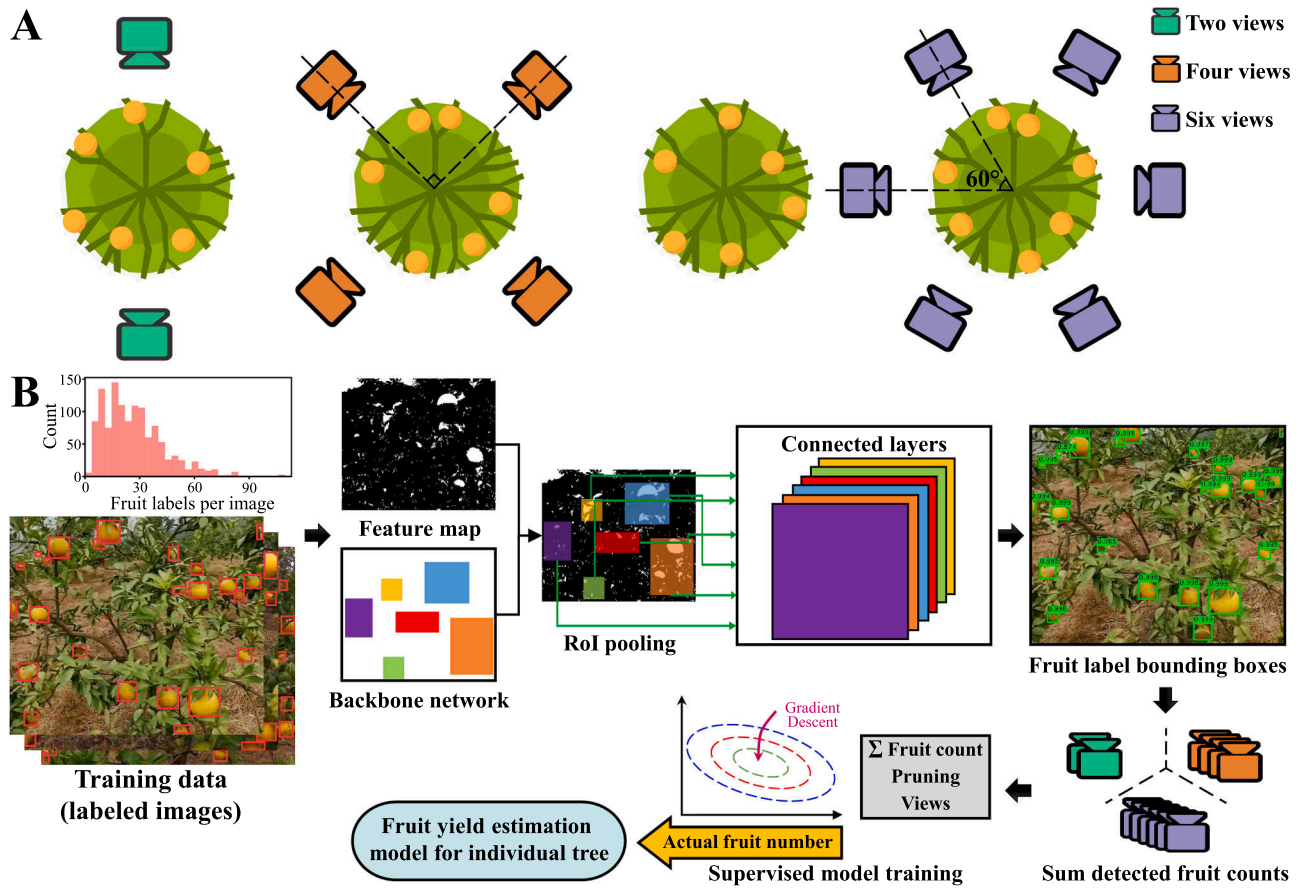


Fig. 1. (A) Illustration of acquiring tree images (part by Vecteezy.com); (B) Procedures of counting fruit and estimating yield by machine learning models.

Table 1

Citrus tree width and the percentage of tree width in the image.

Pruning	Tree width / m	Percentage of tree width in the image / %
None	2.67 ± 0.58	89.38 % - 116.52 %
0–5 %	2.26 ± 0.48	80.04 % - 105.86 %
5–10 %	1.81 ± 0.37	71.88 % - 91.59 %
10–15 %	2.11 ± 0.35	79.57 % - 98.90 %

Adaptive moment estimation was established as the optimization algorithm ($\beta_1=0.900, \beta_2=0.999, \epsilon=10^{-8}$) to improve the convergence speed.

YOLOv5 and YOLOv6 based fruit detection model was trained and tested on the same dataset. The architecture of YOLOv5 and YOLOv6 integrates CSPDarkNet (Center and Scale Prediction) and EfficientRep as the backbone network, respectively, to optimize the feature extraction process. PAN (Path Aggregation Network) and Rep-PAN are as the neck network, respectively, to enhance feature fusion from different layers, thereby contributing to the model's robustness (Wang and He, 2021). To ensure the models maintained a compact footprint suitable for deployment while still providing a high standard for comparison in terms of detection performance, the smallest variant of YOLOv5 and YOLOv6 was utilized and the model parameters (number of iterations, batch size, loss function, etc.) were initialized the same as those of Faster R-CNN. The entire process of these models is illustrated in Fig. 1B.

F1-score was used to evaluate the fruit detection model, it is defined as follow:

$$P = \frac{TP}{TP + FP} \quad (1)$$

$$R = \frac{TP}{TP + FN} \quad (2)$$

$$F1\text{-score} = \frac{2 \times P \times R}{P + R} \quad (3)$$

where P is Precision, R is Recall, TP is true positive, FP is false positive (wrong detection), and FN is false negative (missing detection).

2.3. Yield estimation using machine learning

The detected fruit counts in each type of image views were summed as the fruit count of an individual tree (Fig. 1B). Since missed and overcounting might have occurred during fruit detection, and factors like pruning, image views, tree height and width could make influences, the yield of citrus fruit and the fruit count of individual trees by the fruit detection model are not consistent with a generalized linear model. Therefore, the citrus fruit yield of individual trees was estimated using three supervised machine learning models, including random forest (RF) (Zhu, Chen et al., 2022), support vector machine (SVM) (Tu et al., 2018) and extreme gradient boosting (XGBoost) (Liu et al., 2021). A generalized linear model (GLM) was applied as the benchmark due to its simplicity and interpretability. It implemented maximum likelihood estimation to estimate parameters for the linear regression model. The performance improvements of machine learning models in yield estimation can be effectively evaluated by establishing a baseline with GLM (Niku et al., 2019). All three models are appropriate for modeling data regression with potential non-linear correlations (Guo et al., 2021).

The Boruta algorithm (Kursa and Rudnicki, 2010) was first utilized to test the importance of each independent variable, including pruning intensity, image views, tree height and width. It is an iterative feature

selection algorithm that uses RF to determine feature importance scores iteratively, and compares those scores to shadow features (shuffled copies of all features) to determine which original features are truly important. Then these machine learning models were trained and tested with the important independent variables determined by the Boruta algorithm. The ground truth for model training and statistical analysis was derived from manual counts of the fruits on individual trees, which served as the dependent variable. This manual counting was performed meticulously to ensure they accurately reflected the actual fruit yield. Grid tuning and 10-fold cross-validation were conducted to optimize the model parameters. The input and output of the fruit detection model and the machine learning models are listed in Table 2.

2.4. Statistical analysis

The differences between estimation by the models and the ground truth were evaluated by the percentage error (PE), mean absolute percentage error (MAPE), symmetric mean absolute percentage error (SMAPE) and coefficient of determination (R^2), which are defined as follows:

$$PE = \frac{y_i - x_i}{x_i} \quad (4)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n |PE| \quad (5)$$

$$SMAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - x_i|}{(|y_i| + |x_i|)/2} \quad (6)$$

$$R^2 = \left[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \right]^2 / \left[\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2 \right] \quad (7)$$

where n represents the number of values; x_i represents the true values, y_i represents the values estimated by the model, and $\bar{x}$, $\bar{y}$ represent the mean of true and estimated values, respectively. All the calculations and statistical analyses (analysis of variance) were conducted using the R programming language (version 4.2.1). The *Boruta* (version 7.0.0) R package performed the Boruta algorithm before model training. The *randomForest* (version 4.7–1.1), *e1071* (version 1.7–11) and *xgboost* (version 1.6.0.1) R packages performed model training and testing for RF, SVM and XGBoost, respectively. The *caret* (version 6.0–93) R package performed GLM, parameter grid tuning and 10-fold cross-validation. All the training and testing were conducted on an Intel Core i7 computer with 3.20 GHz and 8 GB of memory. The OpenCV image processing library, version 4.5.1, was used to draw and visualize the fruit labels in images.

3. Results

3.1. Influences on fruit detection

In the fruit detection model evaluation, a threshold is often used to determine if a prediction is correct or not. As *F1-score* and time spent on

Table 2

Input and output of the fruit detection models and the machine learning models.

Model	Input	Output
Fast R-CNN / YOLOv5 / YOLOv6	Tree images	Fruit labels, Fruit count of individual trees
XGBoost / RF / SVM / GLM	Fruit count of individual trees, Image views, Pruning intensity, Tree height and width*	Fruit yield estimation of individual trees

*Only the important independent variables determined by the Boruta algorithm are as the input. Here list all possible independent variables.

prediction of three deep learning models under 10 levels of confidence threshold listed in Table 3, data augmentation has improved *F1-score* by 16.48 %, 13.74 % and 9.99 % on Faster R-CNN, YOLOv5 and YOLOv6, respectively. This implies the importance of data augmentation in improving model robustness and accuracy. The YOLOv6 based citrus fruit detection model has a highest *F1-score* of 0.958 in testing set when the threshold is 0.8, while the Faster R-CNN based model has a highest *F1-score* of 0.947 with the same threshold, higher than that of YOLOv5. This indicates that the YOLOv6 based model is the most accurate for citrus fruit detection, confirming a citrus fruit only if the model predicts a probability above 0.8. Also, YOLOv6 based model is faster than Faster R-CNN when detecting fruit from images, according to the average time spent on prediction (Table 3).

The fruit count of the YOLOv6 based detection model is compared with the actual number of fruit on each citrus tree, and the results are shown in Fig. 2A. This suggests that image views and pruning have a significant influence on the detection of fruit. When two images were taken per tree, the accuracy of fruit count decreased with the increase in actual fruit number, regardless of the pruning intensity. Most fruit counts are fewer than the actual fruit number. When four images were taken per tree, the fruit count was quite accurate for 0–10 % pruning trees with $R^2 > 0.94$ and SMAPE < 0.15 . Yet the fruit count error for non-pruned trees was still considerable. When six images were taken per tree, the results of fruit count still had substantial error (SMAPE > 0.24). Most of the trees pruned by 0–15 % had fruit count larger than the actual number fruit of each tree, indicating that the use of six image views may lead to overcounting. Fruit counting errors under different image views were significantly different according to ANOVA ($p < 0.01$). Fig. 2B also shows the same results and suggests that 5–10 % pruning with four image views had the minimum PE (closest to zero).

The influence of pruning was also considerable. Fruit counting errors under different pruning intensities were significantly different according to ANOVA ($p < 0.01$). Citrus trees that had not been pruned tended to generate large errors in the counts of fruit. Pruning > 10 % also resulted in considerable SMAPE. Pruning 0–5 % and 5–10 % with four and six image views had $R^2 > 0.92$ and SMAPE < 0.3 . The fruit count of pruning 0–15 % with four image views had the lowest SMAPE, indicating four image views were appropriate to estimate the yield on individual trees using only the fruit detection model. No significant differences in terms of fruit counting error were identified between pruning 0–15 % with four image views.

Tree height and width had relatively smaller influences than pruning and image views. No significant differences were identified between fruit counting error and tree height or width according to ANOVA ($p < 0.01$). However, the interactions of pruning $\times$ tree height and pruning $\times$ tree width were identified significant. As shown in Fig. 2B, a positive PE indicated that the count of citrus fruit was larger than the actual fruit of

Table 3

F1-score and time spent on prediction of the citrus fruit detection models.

Threshold	Data NOT augmented			Data augmented		
	Faster R-CNN	YOLOv5	YOLOv6	Faster R-CNN	YOLOv5	YOLOv6
0	0.755	0.762	0.802	0.844	0.819	0.853
0.1	0.722	0.807	0.767	0.848	0.820	0.891
0.2	0.710	0.779	0.725	0.859	0.842	0.902
0.3	0.794	0.790	0.765	0.854	0.827	0.884
0.4	0.821	0.793	0.816	0.885	0.856	0.921
0.5	0.759	0.744	0.821	0.869	0.841	0.885
0.6	0.803	0.778	0.765	0.866	0.898	0.892
0.7	0.804	0.808	0.839	0.940	0.919	0.947
0.8	0.813	0.847	0.871	0.947	0.915	0.958
0.9	0.776	0.846	0.827	0.931	0.907	0.934
Time spent (ms/image)	111.96 ± 4.59	69.84 ± 3.49	72.03 ± 3.21	107.85 ± 5.42	70.56 ± 2.56	75.96 ± 3.08

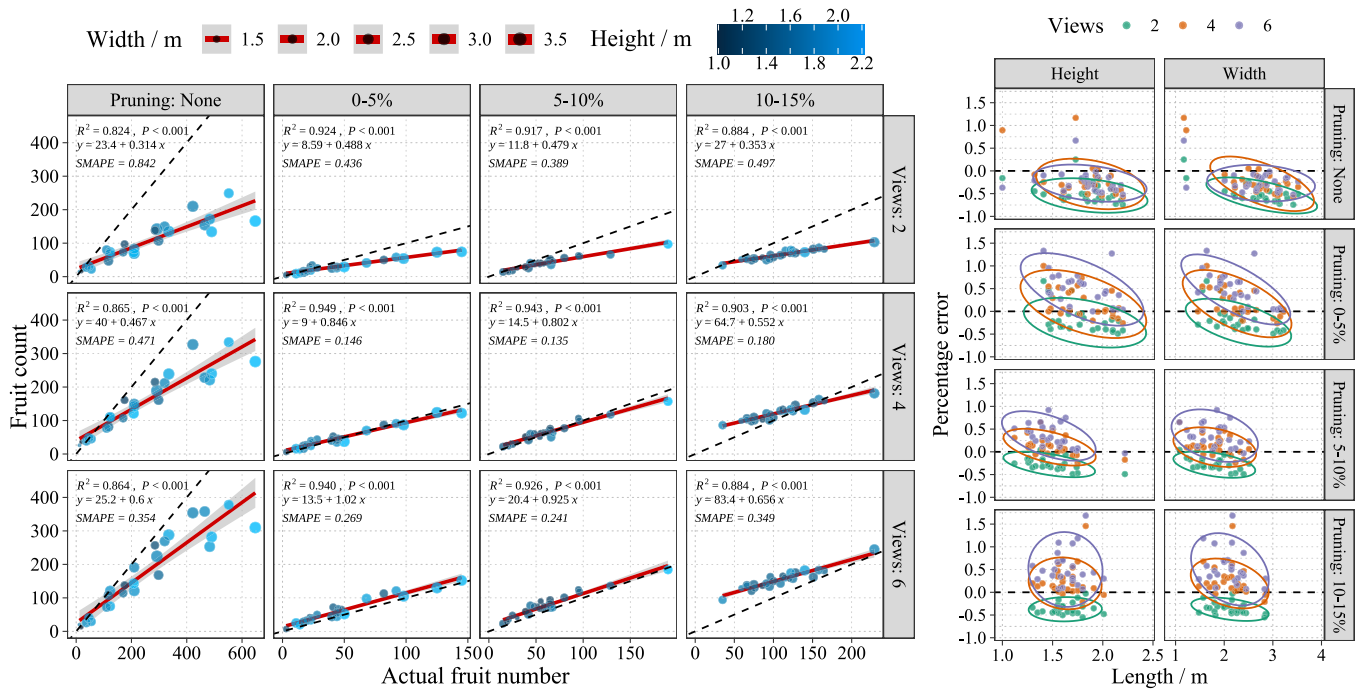


Fig. 2. Performance of the deep learning based citrus fruit detection model. (A) Comparison between the fruit count and actual fruit number of individual trees among different pruning intensities and image views. The dashed lines represent fruit counts that equal the actual fruit number of each tree. The solid lines and gray regions represent the linear regression and 95 % confidence intervals of the scatter plots. The linear regression formula, R^2 and P values are noted on the corresponding sub-plot. (B) Percentage error (PE) of fruit counted among different pruning intensities, tree height and width. The dashed lines represent PE = 0. The ellipses represent the corresponding 95 % confidence intervals.

each tree. For trees that were not pruned, only the shortest trees had a positive PE. As the pruning intensity increased, short and thin trees (with low height and width) had a higher positive PE except the highest PEs appeared in trees with middle-level height and a width under pruning of 10–15 %. As the image views increased, the PE increased and those of the short and thin trees increased more quickly, indicating that taking more images per tree resulted in a larger count of fruit, and overcounting might have occurred under six image views.

3.2. Importance of the variables used to estimate fruit yield

The Boruta algorithm confirmed that all the variables presented above contributed to the estimation of fruit yield. Variable importance was then measured as the percentage decrease in accuracy when a variable was omitted from a standard GLM model. A higher percentage decrease in accuracy indicated that the variable was more important. As shown in Fig. 3, the error bars indicate standard deviation in the results from iterations in the algorithm. Shorter error bars indicate that the results among iterations vary less. Therefore, the importance of each variable is highly convincing. The variable importance (percentage decrease in accuracy) of the five independent variables suggested that

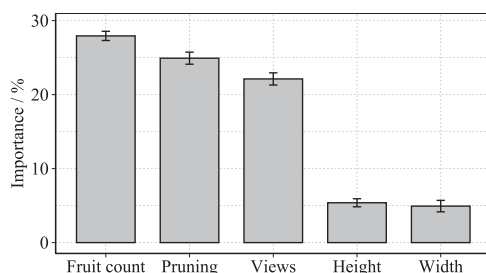


Fig. 3. Variable importance of independent variables to estimate fruit yield.

the fruit count is the most important factor for yield estimation, followed by pruning and image views. Tree height and width were relatively unimportant. Fruit count, pruning and image views were the most important variables. Fruit count was the most crucial and was responsible for 27.94 % importance because it directly correlates with fruit yield. Moreover, pruning, responsible for 24.93 % importance, was highly important for the estimation of fruit yield, suggesting that pruning could have influenced the occlusion of branches and leaves in images. Image views, responsible for 22.12 % importance, were also important because they determined the perspective of investigating a tree. Tree height and width had importance of 5.39 % and 4.93 %, respectively, suggesting they had little impact on fruit counting or the estimation of yield.

3.3. Performance of the model used to estimate fruit yield

Four machine learning models were trained and utilized to estimate the yield of citrus fruit using three variables (fruit count, pruning and image views). Optimizing the parameters is essential for model performance. Thus, the parameters of each machine learning model were grid tuned with five repeats of 10-fold cross-validations for overall estimation error. The overall estimation error of each machine learning model and the corresponding optimal values of parameters are shown in Table 4. The MAPE and SMAPE of the machine learning model training and testing were calculated and evaluated to determine the performance of yield estimation and find the optimal model. SMAPE was utilized to avoid the influence when the actual fruit number is extremely small, which results in an excessively high MAPE value. The process involves >50 iterations of 10-fold cross-validations for each combination of parameters. XGBoost was the optimal machine learning model in terms of the overall MAPE and SMAPE values during both training and testing. The RF errors were small during training, but they were larger than those of XGBoost during testing. SVM models with different kernel

Table 4

Searching optimal parameters for each machine learning model by grid tuning and 10-fold cross-validation.

Model	Parameter	Value	Range	Stage	MAPE	SMAPE
XGBoost	nrounds	110	50–500	training	0.007	0.005
	lambda	1	0, 1	testing	0.171	0.153
	alpha	1	0, 1			
	eta	0.1	10^{-3} - 1			
RF	mtry	5	1–10	training	0.083	0.086
	splitrule	extratrees	multiple†	testing	0.190	0.194
	min.node.size	1	0–5			
	degree	4	2–5	training	0.511	0.252
SVM(POLY)*	C	1000	10^2 - 10^5	testing	0.552	0.241
	scale	no	yes, no			
	sigma	0.02	10^{-3} - 1	training	0.152	0.155
	C	1000	10^2 - 10^5	testing	0.218	0.183
GLM				training	0.597	0.409
				testing	0.706	0.478

*The abbreviation in parentheses refers to the kernel function used in the SVM model. Different kernel functions have different parameters. POLY, polynomial function; RBF, radial based function.

†The splitrule parameter of RF includes: variance, extratrees, and maxstat.

functions had different parameters. SVM with polynomial kernel function has an additional parameter: *degree*, which indicates the degree of the polynomial function. Consequently, results of the best two kernel functions (RBF and polynomial) are investigated and shown separately. SVM with radial based functions produced smaller errors than those with polynomial functions during both training and testing. Although XGBoost had the lowest errors, SVM(RBF) had the lowest gaps of errors between training and testing, indicating that SVM(RBF) is insensitive to the input variables. GLM, as the benchmark, presented the largest MAPE and SMAPE, indicating that these machine learning models outperform linear models. Among the training and testing of all the models, the SMAPE values were not substantially smaller than those of the MAPE. The fact that a few trees having fewer than 10 fruits indicates that extremely small values do not influence the model performance.

Fig. 4 shows the performance of the three optimal machine learning models with optimal key parameters under different levels of pruning intensity and image views. The errors MAPE and SMAPE of the model on the testing sets were calculated and compared. The errors of estimation

under the four image views were the lowest among three levels of image views for all the models. Under two image views, the estimation errors decreased with the increase in pruning intensity. Under four and six image views, the errors decreased and then increased and the lowest errors occurred at pruning 0–10 % for all the three models. The differences between MAPE and SMAPE were negligible in most cases except for the errors of RF and SVM(RBF) under two image views without pruning. This indicates that those extremely small actual numbers of fruit do not influence the results.

When the trees were not pruned, the errors of estimation of the three models were >0.18 under two and four image views, and they were not low under six image views, indicating these machine learning models do not have ideal performance for citrus trees that have not been pruned. When pruning 10–15 %, all three models had the lowest error of estimation under two image views, while there were errors close to or >0.2 under six image views. This suggests that applying two image views is appropriate for citrus trees with new shoots pruned by 10–15 %. Trees that were pruned by 0–10 % did not display any clear trends on the

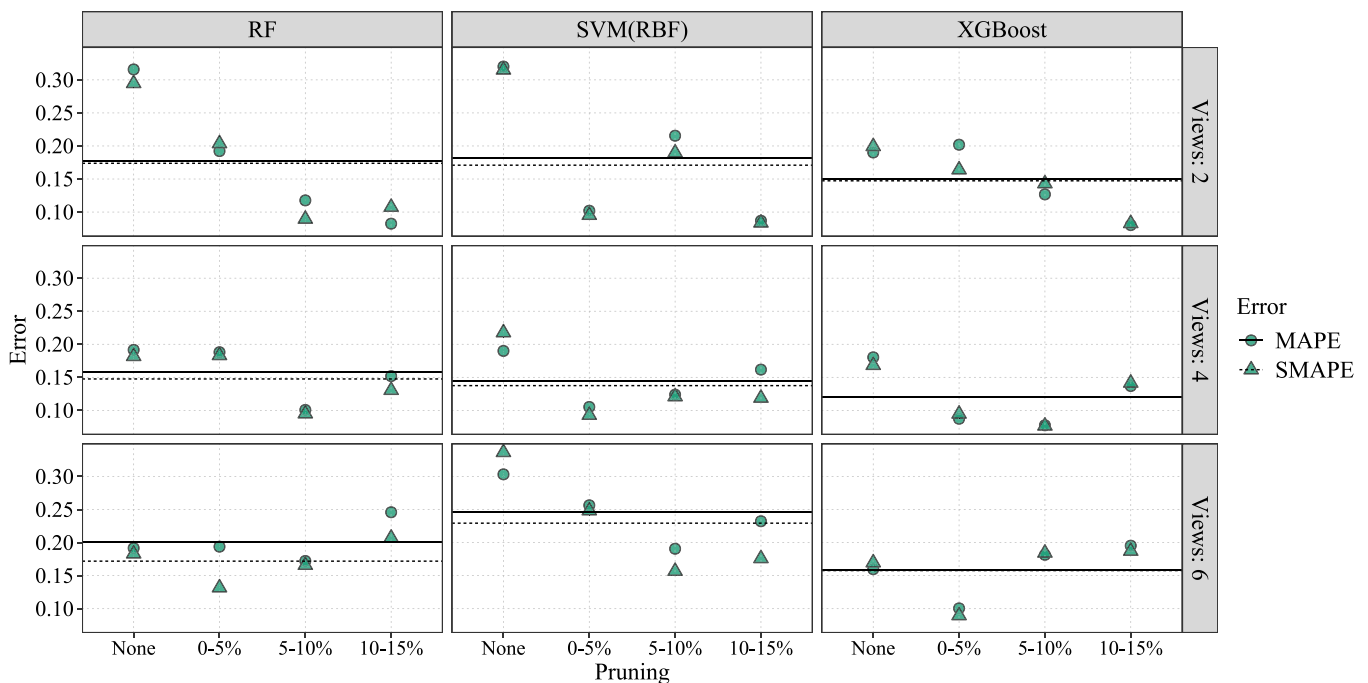


Fig. 4. The testing performance of supervised machine learning models with optimal key parameters in estimating fruit yield of individual trees. The horizontal lines represent the mean of the corresponding error values.

estimation of errors. RF and XGBoost had the lowest errors when the trees had been pruned by 5–10 % and 0–5 % under four and six image views, respectively. In contrast, SVM(RBF) showed the opposite compared with XGBoost. Among all three models, higher pruning intensity or more image views did not reduce the errors in estimating yields, suggesting that appropriate image views are required to accurately estimate the yield under different pruning intensities. Considering the errors in estimating fruit yield and the gaps between training and testing under different pruning and image views, XGBoost had the best model performance for non-pruned and pruned by ≤ 5 % under six views, 5–10 % pruning under four views, and 10–15 % pruning under two views.

4. Discussion

The effects of pruning and imaging views were examined through various treatments. Our dataset of citrus tree images, including different pruning intensities and multiple angles, allows us to enhance the model's robustness and accuracy in fruit detection. Thus, the influences of pruning and imaging views have been examined by different treatments. Additionally, we train machine learning models to assess the relationship between the predicted fruit counts and the actual fruit counts. By systematically validating the model's outputs against the manual counts, we ensure that the recognized fruits from the images accurately reflected the real-world fruit yield. This approach addresses the challenge of correlating image-based detections with actual fruit presence on trees, thereby enhancing the reliability of yield estimation.

Ground imaging systems in citrus orchards are crucial for fruit detection, typically imaging from two opposite sides (Zhuang et al., 2018). This image view offers adequate accuracy in fruit detection, especially when the focus is on overall orchard yield. Trees are often pruned to expose the inner canopy, which aligns with findings that a 10–15 % pruning level minimizes estimation errors under two image views. However, pruning primarily aims to boost yield and prevent disease, not specifically to enhance yield estimation. Over-pruning can actually reduce fruit yield (Chueca et al., 2021). To improve yield estimation, more combinations of image views and pruning intensity are tested. The results suggest that four and six image views per tree are optimal for estimating the fruit yield of trees pruned by ≤ 10 % and those not pruned, respectively. This strategy ensures more accurate fruit yield estimation without compromising tree health or yield.

Pruning impacts fruit yield estimation potentially due to changes in canopy architecture (Huang et al., 2022). Consequently, the study examines how pruning influences the accuracy of yield estimation. The fruit counting result without machine learning models (RF, SVM, and XGBoost) and the yield estimation of these models suggest that increased pruning intensity reduces the number of image views needed for optimal yield estimation. This is likely because more severe pruning results in fewer branches and leaves, making fruits more visible (as shown in

Fig. 5). Moreover, as pruning intensity and the number of image views increase, particularly at six image views, the fruit count may exceed the actual count (Fig. 2A). This suggests that overcounting could occur with more than four image views or when trees are pruned by >10 %, in consistent with additional research (Dorj et al., 2017).

Tree height and width decrease as the PE increases among all levels of pruning intensity (Fig. 2B) due to less severe pruning allowing more new shoots to grow, some of which develop into vertical upward branches. Despite these changes, the camera's adjusted height and coverage of the canopy largely negate the influence of tree height and width on detection results. The Boruta analysis, which evaluates and ranks the variables in the order of importance (Koren et al., 2021), has indicated that tree height and width contribute only about 5 % to the yield estimation model. Such conclusion has rarely presented in previous studies because they have primarily focused on the fruit yield of an area or a block in the orchard (Lee et al., 2020; Jiang et al., 2023), instead of the yield of individual trees or plants.

Machine learning models are gaining popularity for estimating or predicting fruit yield, but no single model has been found universally applicable to all cases of citrus trees (Maheswari et al., 2021; He et al., 2022). The SVM(RBF) model is noted for its robustness, with the smallest estimation error gaps between training and testing. However, both RF and XGBoost models have lower estimation errors, with XGBoost showing the lowest errors under specific pruning and imaging conditions. Nevertheless, the XGBoost model outperformed others, which aligns with previous research (Obsie et al., 2020).

The combination of the YOLOv6 for fruit detection and XGBoost for yield estimation has been deemed effective for individual citrus trees. This accords with previous research suggesting combination of machine learning models tends to outperform single models (Elavarasan and Durai Raj Vincent, 2021). The optimal number of ground-based image views recommended are two for trees pruned >10 %, four for 5–10 % pruning, and six for ≤ 5 % pruning. For most cases with 5–10 % pruning, both two and four image views are acceptable for fruit yield estimation.

In current citrus cultivation practices, pruning aims to make citrus fruits more visible on the canopy, making foliage and canopy structure the primary influence on fruit detection in productive scenarios (Matias et al., 2023). Also, orchard block layouts with compact rows and columns also pose challenges for ground imaging systems, necessitating the use of wide-angle lenses. While this study demonstrates the potential of combining deep learning and machine learning for citrus yield estimation, its generalizability may be limited due to the specific cultivar and location used, reliance on ground-based imaging, and the representativeness of the datasets. Furthermore, other influential factors such as soil quality and pest pressures were not considered. Despite these challenges, the results in this study suggest that the proposed methods are viable for estimating citrus fruit yield for individual trees, offering a promising approach to enhance yield estimation accuracy and efficiency in citrus cultivation.

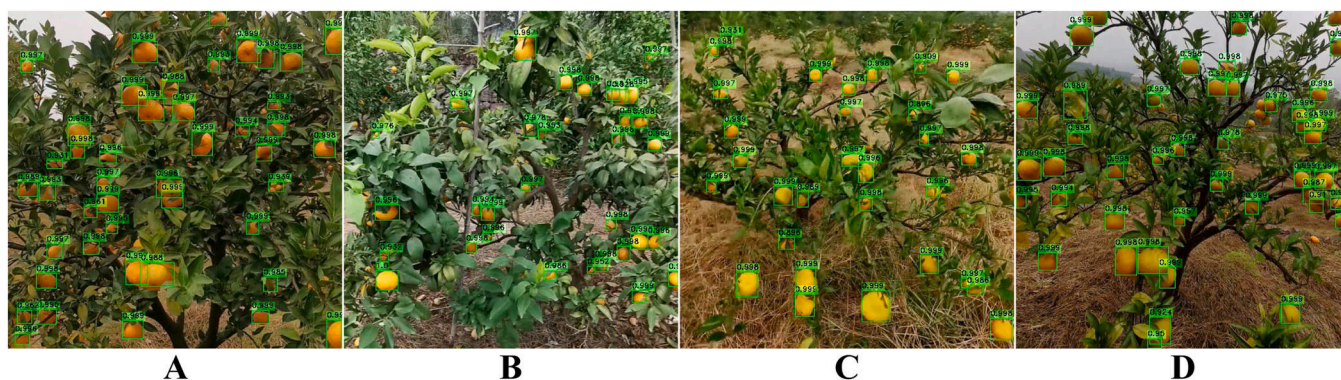


Fig. 5. Citrus fruit detection on trees with different levels of pruning intensity. (A) none. (B) 0–5 %. (C) 5–10 %. (D) 10–15 %.

5. Conclusions

This study explored the potential of a method that combined deep learning, machine learning and a ground imaging system to estimate the citrus yields of individual trees. A YOLOv6 based detection model has been built to accurately detect the fruit on citrus tree images and outperforms the benchmark, which is in accord with previous studies (Apolo-Apolo et al., 2020). Our analysis revealed that fruit count, pruning intensity and image views significantly influence fruit yield estimation, while tree height and width have negligible impact. Specifically, the XGBoost model demonstrated the lowest estimation errors under optimal conditions: two, four, and six image views for trees pruned by >10 %, 5–10 %, and ≤5 %, respectively. The MAPE and SMAPE values indicated that these combinations provided the most accurate yield estimates. The results demonstrate the applicability of this approach to accurately estimate citrus fruit yield on individual trees, with the potential to enhance precision orchard management and optimize resource allocation based on reliable yield predictions.

CRediT authorship contribution statement

Qing Gu: Methodology, Formal analysis. **Xiaobin Zhang:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Funding acquisition, Conceptualization. **Feng Liu:** Resources, Methodology, Investigation. **Yiying Zhao:** Methodology, Investigation. **Yihang Zhu:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgements

The work was supported by the Crop Phenotype High-Throughput Acquisition and Intelligent Application Platform Projects [10410150123CH2901G]; the Key Scientific and Technological Grant of Zhejiang for Breeding New Agricultural Varieties [2021C02066-1] and the Ningbo Key Laboratory of Characteristic Horticultural Crops in Quality Adjustment and Resistance Breeding [NBYYL2023009].

References

- Apolo-Apolo, O.E., Martínez-Guanter, J., Egea, G., Raja, P., Pérez-Ruiz, M., 2020. Deep learning techniques for estimation of the yield and size of citrus fruits using a UAV. *Eur. J. Agron.* 115, 126030 <https://doi.org/10.1016/j.eja.2020.126030>.
- Budiarto, R., Poerwanto, R., Santosa, E., Efendi, D., Agustina, A., 2022. Comparative and correlation analysis of young and mature kaffir lime (*Citrus hystrix* DC) leaf characteristics. *Int. J. Plant Biol.* 13 (3), 270–280. <https://doi.org/10.3390/ijpb13030023>.
- Chen, C., Wang, L., Liu, H., Liu, J., Xu, W., Huang, M., Gou, N., Wang, C., Bai, H., Jia, G., Wuyun, T., 2024. Construction of apricot variety search engine based on deep learning. *Hortic. Plant J.* 10, 387–397. <https://doi.org/10.1016/j.hpj.2023.02.007>.
- Chen, Y., An, X., Gao, S., Li, S., Kang, H., 2021. A deep learning-based vision system combining detection and tracking for fast on-line citrus sorting. *Front. Plant Sci.* 12 <https://doi.org/10.3389/fpls.2021.622062>.
- Chueca, P., Mateu, G., Garcerá, C., Fonte, A., Ortiz, C., Torregrosa, A., 2021. Yield and economic results of different mechanical pruning strategies on “Navel Foyos” oranges in the mediterranean area. *Agriculture* 11 (1), 82. <https://doi.org/10.3390/agriculture11010082>.

- Costa, L., Kunwar, S., Ampatzidis, Y., Albrecht, U., 2022. Determining leaf nutrient concentrations in citrus trees using UAV imagery and machine learning. *Precis. Agric.* 23 (3), 854–875. <https://doi.org/10.1007/s11119-021-09864-1>.
- de Luna, R.G., Dadios, E.P., Bandala, A.A., Viceria, R.R.P., 2020. Tomato growth stage monitoring for smart farm using deep transfer learning with machine learning-based maturity grading. *AGRIVITA J. Agric. Sci.* 42 (1) <https://doi.org/10.17503/agrivita.v42i1.2499>.
- Dorj, U., Lee, M., Yun, S., 2017. An yield estimation in citrus orchards via fruit detection and counting using image processing. *Comput. Electron. Agr.* 140, 103–112. <https://doi.org/10.1016/j.compag.2017.05.019>.
- Elavarasan, D., Durai Raj Vincent, P.M., 2021. Fuzzy deep learning-based crop yield prediction model for sustainable agronomical frameworks. *Neural Comput. Appl.* 33 (20), 13205–13224. <https://doi.org/10.1007/s00521-021-05950-7>.
- Gao, F., Fu, L., Zhang, X., Majeed, Y., Li, R., Karkee, M., Zhang, Q., 2020. Multi-class fruit-on-plant detection for apple in SNAP system using Faster R-CNN. *Comput. Electron. Agr.* 176, 105634 <https://doi.org/10.1016/j.compag.2020.105634>.
- Gongal, A., Amatya, S., Karkee, M., Zhang, Q., Lewis, K., 2015. Sensors and systems for fruit detection and localization: a review. *Comput. Electron. Agr.* 116, 8–19. <https://doi.org/10.1016/j.compag.2015.05.021>.
- Guo, Y., Y. Fu, F. Hao, X. Zhang, W. Wu, X. Jin, C. Robin Bryant and J. Senthilnath. 2021. Integrated phenology and climate in rice yields prediction using machine learning methods. *Ecol. Indic.* 120: <https://doi.org/10.1016/j.ecolind.2020.106935>.
- He, L., Fang, W., Zhao, G., Wu, Z., Fu, L., Li, R., Majeed, Y., Dhupia, J., 2022. Fruit yield prediction and estimation in orchards: a state-of-the-art comprehensive review for both direct and indirect methods. *Comput. Electron. Agr.* 195, 106812 <https://doi.org/10.1016/j.compag.2022.106812>.
- Huang, H., Huang, T., Li, Z., Lyu, S., Hong, T., 2022. Design of citrus fruit detection system based on mobile platform and edge computer device. *Sensors* 22 (1), 59. <https://doi.org/10.3390/s22010059>.
- Jiang, X., Wu, Z., Han, S., Yan, H., Zhou, B., Li, J., 2023. A multi-scale approach to detecting standing dead trees in UAV RGB images based on improved faster R-CNN. *PLOS One* 18, e0281084. <https://doi.org/10.1371/journal.pone.0281084>.
- Karydas, C., Iatrou, M., Kouretas, D., Patouna, A., Iatrou, G., Lazos, N., Gewehr, S., Tseni, X., Tekos, F., Zartaloudis, Z., Mainos, E., Mourelatos, S., 2020. Prediction of antioxidant activity of cherry fruits from uas multispectral imagery using machine learning. *Antioxidants* 9 (2), 156. <https://doi.org/10.3390/antiox9020156>.
- Khattak, A., Asghar, M.U., Batool, U., Asghar, M.Z., Ullah, H., Al-Rakhami, M., Gumaei, A., 2021. Automatic detection of citrus fruit and leaves diseases using deep neural network model. *IEEE Access* 9, 112942–112954. <https://doi.org/10.1109/ACCESS.2021.3096895>.
- Koren, M., Jakuš, R., Zápotocký, M., Barka, I., Holuša, J., Đuračiová, R., Blaženec, M., 2021. Assessment of machine learning algorithms for modeling the spatial distribution of bark beetle infestation. *Forests* 12 (4), 395. <https://doi.org/10.3390/f12040395>.
- Kursa, M.B., Rudnicki, W.R., 2010. Feature selection with the boruta package. *J. Stat. Softw.* 36 (11) <https://doi.org/10.18637/jss.v036.i11>.
- Lee, M.A., Monteiro, A., Barclay, A., Marcar, J., Miteva-Neagu, M., Parker, J., 2020. A framework for predicting soft-fruit yields and phenology using embedded, networked microprocessors, coupled weather models and machine-learning techniques. *Comput. Electron. Agr.* 168, 105103 <https://doi.org/10.1016/j.compag.2019.105103>.
- Liu, G., Chen, Y., He, X., Yao, F., Guan, G., Zhong, B., Zhou, G., 2020. Seasonal changes of mineral nutrients in the fruit of navel orange plants grafted on trifoliate orange and citrange. *Sci. Hortic. -Amst.* 264, 109156 <https://doi.org/10.1016/j.scienta.2019.109156>.
- Liu, Y., Wang, H., Fei, Y., Liu, Y., Shen, L., Zhuang, Z., Zhang, X., 2021. Research on the prediction of green plum acidity based on improved XGBoost. *Sensors* 21 (3), 930. <https://doi.org/10.3390/s21030930>.
- Maheswari, P., Raja, P., Apolo-Apolo, O.E., Pérez-Ruiz, M., 2021. Intelligent fruit yield estimation for orchards using deep learning based semantic segmentation techniques—a review. *Front. Plant Sci.* 12 <https://doi.org/10.3389/fpls.2021.684328>.
- Mai, X., Zhang, H., Jia, X., Meng, M.Q.-H., 2020. Faster R-CNN With classifier fusion for automatic detection of small fruits. *IEEE Trans. Autom. Sci. Eng.* 17, 1–15. <https://doi.org/10.1109/TASE.2020.2964289>.
- Matias, P., Barrote, I., Azinheira, G., Continella, A., Duarte, A., 2023. Citrus pruning in the mediterranean climate: a review. *Plants* 12, 3360. <https://doi.org/10.3390/plants12193360>.
- Mazzia, V., Khaliq, A., Salvetti, F., Chiaberge, M., 2020. Real-time apple detection system using embedded systems with hardware accelerators: an edge AI application. *IEEE Access* 8, 9102–9114. <https://doi.org/10.1109/ACCESS.2020.2964608>.
- Moriya, É.A.S., Imai, N.N., Tommaselli, A.M.G., Berveglieri, A., Santos, G.H., Soares, M. A., Marino, M., Reis, T.T., 2021. Detection and mapping of trees infected with citrus gummosis using UAV hyperspectral data. *Comput. Electron. Agr.* 188, 106298 <https://doi.org/10.1016/j.compag.2021.106298>.
- Niku, J., Brooks, W., Herliansyah, R., Hui, F.K.C., Taskinen, S., Warton, D.I., 2019. Efficient estimation of generalized linear latent variable models. *PLOS One* 14, e0216129. <https://doi.org/10.1371/journal.pone.0216129>.
- Obsie, E.Y., Qu, H., Drummond, F., 2020. Wild blueberry yield prediction using a combination of computer simulation and machine learning algorithms. *Comput. Electron. Agr.* 178, 105778 <https://doi.org/10.1016/j.compag.2020.105778>.
- Ojo, I.A., L. Costa, Y. Ampatzidis, F. Alferez and S. Shukla. 2021. Citrus Fruit Maturity Prediction Utilizing UAV Multispectral Imaging and Machine Learning. 2021 ASABE Annual International Virtual Meeting. St. Joseph, MI, ASABE: 1. <https://doi.org/10.13031/aim.202100495>.

- Radcliffe, J., Cox, J., Bulanon, D.M., 2018. Machine vision for orchard navigation. *Comput. Ind.* 98, 165–171. <https://doi.org/10.1016/j.compind.2018.03.008>.
- Tu, Y., Bian, M., Wan, Y., Fei, T., 2018. Tea cultivar classification and biochemical parameter estimation from hyperspectral imagery obtained by UAV. *PeerJ* 6, e4858. <https://doi.org/10.7717/peerj.4858>.
- Wang, D., He, D., 2021. Channel pruned YOLO V5s-based deep learning approach for rapid and accurate apple fruitlet detection before fruit thinning. *Biosyst. Eng.* 210, 271–281. <https://doi.org/10.1016/j.biosystemseng.2021.08.015>.
- Xin, Y., Wu, Y., Qiao, B., Su, L., Xie, S., Ling, P., 2022. Evaluation on the phenotypic diversity of Calamansi (*Citrus microcarpa*) germplasm in Hainan island. *Sci. Rep.* 12, 371. <https://doi.org/10.1038/s41598-021-03775-x>.
- Xu, D., Zhao, H., Lawal, O.M., Lu, X., Ren, R., Zhang, S., 2023. An automatic jujube fruit detection and ripeness inspection method in the natural environment. *Agronomy* 13, 451. <https://doi.org/10.3390/agronomy13020451>.
- Zeeshan, S., Aized, T., Riaz, F., 2023. The design and evaluation of an orange-fruit detection model in a dynamic environment using a convolutional neural network. *Sustainability* 15, 4329. <https://doi.org/10.3390/su15054329>.
- Zhang, W., Wang, J., Liu, Y., Chen, K., Li, H., Duan, Y., Wu, W., Shi, Y., Guo, W., 2022. Deep-learning-based in-field citrus fruit detection and tracking. *Hortic. Res.* 9 <https://doi.org/10.1093/hr/uhac003>.
- Zhang, W., Chen, K., Zheng, C., Liu, Y., Guo, W., 2022. EasyDAM_V2: efficient data labeling method for multishape, cross-species fruit detection. *Plant Phenomics* 2022, 9761674. <https://doi.org/10.34133/2022/9761674>.
- Zheng, Z., Xiong, J., Lin, H., Han, Y., Sun, B., Xie, Z., Yang, Z., Wang, C., 2021. A method of green citrus detection in natural environments using a deep convolutional neural network. *Front. Plant Sci.* 12 <https://doi.org/10.3389/fpls.2021.705737>.
- Zhu, Y., Chen, M., Gu, Q., Zhao, Y., Zhang, X., Sun, Q., Gu, X., Zheng, K., 2022. Machine learning methods for efficient and automated in situ monitoring of peach flowering phenology. *Comput. Electron. Agr.* 202, 107370 <https://doi.org/10.1016/j.compag.2022.107370>.
- Zhu, Y., Gu, Q., Zhao, Y., Wan, H., Wang, R., Zhang, X., Cheng, Y., 2022. Quantitative extraction and evaluation of tomato fruit phenotypes based on image recognition. *Front. Plant Sci.* 13, 859290 <https://doi.org/10.3389/fpls.2022.859290>.
- Zhuang, J.J., Luo, S.M., Hou, C.J., Tang, Y., He, Y., Xue, X.Y., 2018. Detection of orchard citrus fruits using a monocular machine vision-based method for automatic fruit picking applications. *Comput. Electron. Agr.* 152, 64–73. <https://doi.org/10.1016/j.compag.2018.07.004>.